

TECHNO INDIA COLLEGE OF TECHNOLOGY

Department of Computer Science & Engineering

Computer Network Lab Manual

Paper Code: CS692



Objective: Understand the operation of various mechanisms/protocols that operate at network layer: IP fragmentation, DHCP

LAB 7

Demonstration of IP fragmentation & DHCP using Wireshark

3
Hours

Experiment 1:

IP Fragmentation

Mr. X saw the IP packet format and is particularly fascinated by the fragmentation fields and wants to see it in action. Design an experiment where 1) No fragmentation occurs and 2) Fragmentation occurs.

Guidance:

1. You are yet to cover socket programming. “sendUDP.c” is a simple socket program (provided in the directory) that generates a single IP packet of a given size and sends it to the specified destination IP address. Compile the program and design your experiment around it.
2. Which destination IP address should you use? Try both an existing host and non-existent host within subnet and see what happens.

Questions:

Answer the following questions in your report.

1. Specify the exact commands used to conduct the experiment.
2. For the case when no fragmentation happened, note down the values corresponding to the following fields: Total length, Identifier, flags and fragment offset. Why is the total length field so?
3. For the case when fragmentation happened, for each fragment, note down the values corresponding to the following fields: Total length, Identifier, flags and fragment offset. Do the values make sense based on what was covered in theory?

Experiment 2:

Dynamic Host Configuration Protocol (DHCP)

Mr. X is equally fascinated by how hosts obtain IP address and wants to look at the message exchange of this process. One of his staff already procured such a trace, help him interpret the trace.

Guidance:

1. Configuring IP addresses requires root permission. Since you do not have these privileges, for this exercise you will have to make do with a generated trace file “dhcp.out”. This trace file was generated using tcpdump and running “dhclient eth0” on a terminal with root permissions. Explore it via wireshark.

Questions:

Answer the following questions

1. What is the IP address of the DHCP server and on what port is it listening on?
2. Was any DHCP relay involved in forwarding the DHCP packets? How did you determine the answer?
3. When the DHCP server replied, which IP address did it reply to? And why?
4. What is the offered IP address to the client and for how long is this address valid?
5. Apart from the IP address, what additional information has the client received from the dhcp server?